

Raindrop Size Distribution and Sampling Size Errors

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ABSTRACT

The standard deviation of the rain intensity R and the reflectivity factor Z are derived theoretically for R and Z values which are calculated from measured distributions of raindrop size. The derivation is based on the assumption that the distribution of raindrop size follows a negative exponential law. The Poisson distribution is assumed for the number of the counted drops with diameters between D and $D+dD$. We found that a large sample is necessary to get a good estimate of R and Z ; for example, in a widespread rain with a rainfall rate of 1 mm hr^{-1} , a filter paper with an area of $\sim 1 \text{ m}^2$ must be exposed during 1 sec to obtain, with a probability of 68%, a Z value which deviates less than 20% from the mean.

1. Introduction

Many recent measurements of the distributions of raindrop size have been made to evaluate the reflectivity factor Z and the rainfall rate R for use in the Z - R relationship (e.g., Lamp, 1958; Fujiwara, 1965; Stout and Mueller, 1968). The parameters a and b of the Z - R relation

$$Z = aR^b, \quad (1)$$

where Z is measured in $\text{mm}^6 \text{ m}^{-3}$ and R in mm hr^{-1} show considerable variations, even during the same rainfall. The distributions of raindrops can be measured in two ways: either by sampling the drops having fallen during the time t on a surface of an area F (filter paper), or by sampling the drops in an illuminated volume V at a given moment (drop camera).

The larger the sample the more representative are the values. Normally, a filter with a surface $F = 0.01 \text{ m}^2$ is exposed during 1 sec. As we shall see, this sample of $Ft = 0.01 \text{ m}^2 \text{ sec}$ is too small to get statistically meaningful values of R and Z .

The purpose of this paper is to obtain a rough estimate of the standard deviation of the quantities R and Z due to limited sample size. Knowing the standard deviation, we are then able to estimate whether a variation in the distribution of the raindrops or of the Z - R relationship is caused by a change in the meteorological situation or by the statistical fluctuations of the raindrops.

2. Assumptions

The raindrop size distribution is often found to follow the negative exponential law (first proposed by Marshall and Palmer, 1948)

$$N_{(D)} = N_0 e^{-\lambda D}, \quad (2)$$

where $N_{(D)}$ represents the number of drops per millimeter diameter interval and per cubic meter of air, N_0 is a constant and λ a parameter depending on the rainfall rate R . Table 1 gives typical average values of N_0 and

λ for different types of rain (Joss *et al.*, 1968) as well as a specific value obtained by Marshall and Palmer (1948).

For a certain type of rainfall and a certain rainfall rate R one finds a largest drop with a diameter $D_{\max}$ (see, e.g., Diem 1966). We therefore approximated the distribution of the raindrops by a truncated exponential law:

$$\left. \begin{aligned} N_{(D)} &= N_0 e^{-\lambda D} & \text{if } D < D_{\max} \\ N_{(D)} &= 0 & \text{if } D > D_{\max} \end{aligned} \right\}. \quad (3)$$

To simplify the calculations we assumed a single value of $D_{\max}$ equal to 6 mm. We assumed that the numbers $n_1, n_2, n_3, \dots$ of the drops, with diameters between D_n and $D_n + dD$, counted in different volumes $V_1, V_2, V_3, \dots$ of the same size, are distributed according to the Poisson distribution. Each volume V_i represents a sample of the same meteorological situation. The probability $p(n_i)$ of finding n_i drops with diameters between D_n and $D_n + dD$ in one cube V_i is

$$p(n_i) = \frac{(\bar{n})^{n_i}}{(n_i)!} e^{-\bar{n}}. \quad (4)$$

The assumption of the Poisson distribution is confirmed by theoretical and experimental investigations carried out by Sasyo (1965). Assuming the Poisson distribution with the mean value $\bar{n}$, we know the variance σ_n^2 and the standard deviation σ_n for the numbers $n_1, n_2, n_3, \dots$, i.e.,

$$\left. \begin{aligned} \sigma_n^2 &= \bar{n} \\ \sigma_n &= (\bar{n})^{1/2} \end{aligned} \right\}. \quad (5)$$

According to Eq. (2) we assume the mean value $\bar{n}$ to be

$$\bar{n} = V N_0 e^{-\lambda D_n} dD. \quad (6)$$

3. Calculations

Knowing the variance σ_n^2 of the number n of drops with diameters between D_n and $D_n + dD$ in the volume

TABLE 1. N_0 and λ values for different types of rainfall.

Type of rainfall	N_0 [$\text{m}^{-3} \text{mm}^{-1}$]	λ - R relationship [λ (mm^{-1}) R (mm hr^{-1})]	Case
Drizzle	30,000	$\lambda = 5.7R^{-0.21}$	a
Widespread	7000	$\lambda = 4.1R^{-0.21}$	b
Thunderstorm	1400	$\lambda = 3.0R^{-0.21}$	c
Marshall and Palmer	8000	$\lambda = 4.1R^{-0.21}$	d

V , we are able to calculate the variance $\sigma_{n'+n''}^2$ of drops with diameters between $D_{n'}$ and $D_{n'}+dD$ and between $D_{n''}$ and $D_{n''}+dD$ in the volume V due to the addition theorem of variances; thus

$$\sigma_{n'+n''}^2 = \sigma_{n'}^2 + \sigma_{n''}^2 = \bar{n}' + \bar{n}'', \quad (7)$$

assuming the two numbers n' and n'' to be stochastically independent. The variance σ_N^2 of the total number N of drops in the volume V is calculated in a similar manner by using Eqs. (6) and (3) and replacing the sum by the integral. Thus, we obtain

$$\sigma_N^2 = \int_0^{D_{\max}} VN_0 e^{-\lambda D} dD = \frac{N_0 V}{\lambda} \int_0^{X_{\max}} e^{-X} dX$$

$$= \frac{N_0 V}{\lambda} \gamma(1, X_{\max}), \quad (8)$$

where $X = \lambda D$, $X_{\max} = \lambda D_{\max}$ and $\gamma(\alpha, X_{\max})$ is the incomplete Gamma function which is related to the complete Gamma function $\Gamma(\alpha)$ by

$$\gamma(\alpha, \infty) = \Gamma(\alpha) = (\alpha-1)!$$

We now find the mean total number of drops $\bar{N}$ in the volume V from Eq. (6), using the addition theorem of mean values; thus,

$$\bar{N} = \int_0^{D_{\max}} VN_0 e^{-\lambda D} dD = \frac{N_0 V}{\lambda} \int_0^{X_{\max}} e^{-X} dX$$

$$= \frac{N_0 V}{\lambda} \gamma(1, X_{\max}), \quad (9)$$

where from (8) we find that:

$$\bar{N} = \sigma_N^2, \quad (10)$$

or

$$\frac{\sigma_N}{\bar{N}} = (\bar{N})^{-1/2}. \quad (11)$$

By using a Poisson distribution of the total number of drops N in the sample, we have shown (as might be expected) that the addition theorem in such a distribution, assuming stochastic independence of drops with different diameters, gives rises to an equality between $\bar{N}$ and σ_N^2 . If we now approximate the Poisson distribution by the normal distribution (see, e.g., Hald,

1952), we find that the total number N of drops in the sample may be assumed to be approximately normally distributed about the mean $\bar{N}$ with variance $\bar{N}$ for large values of $\bar{N}$; i.e., there is a probability of $\sim 32\%$ of finding a value of N deviating more than σ_N from the mean value $\bar{N}$, or a probability P_2 of $\sim 4.5\%$ of finding a N value deviating more than $2\sigma_N$ from the mean value $\bar{N}$. This can be expressed as

$$\left. \begin{aligned} P_1 &= P[N < (\bar{N} - \sigma_N); \quad N > (\bar{N} + \sigma_N)] \cong 32\% \\ P_2 &= P[N < (\bar{N} - 2\sigma_N); \quad N > (\bar{N} + 2\sigma_N)] \cong 4.5\% \end{aligned} \right\}. \quad (12)$$

The following calculations presume the approximation of the Poisson distribution by the normal distribution; this is always allowed if the quotients of σ_N/N , σ_R/R or σ_Z/Z are small, i.e., the sample is large enough to derive meaningful values of N , R or Z .

4. The variance of R and Z

The radar reflectivity z of n drops with a diameter D_n is

$$z = nD_n^6. \quad (13)$$

Knowing the standard deviation σ_n of drops with diameters between D_n and D_n+dD [Eq. (5)], we find that the corresponding standard deviation σ_z and variance σ_z^2 of the radar reflectivity are

$$\sigma_z = \bar{n}^{\frac{1}{2}} D_n^6, \quad (14)$$

$$\sigma_z^2 = \bar{n} D_n^{12}. \quad (15)$$

We now find that the variance σ_Z^2 of the total reflectivity factor in a cube with the volume V is

$$\sigma_Z^2 = \int_0^{D_{\max}} VN_0 e^{-\lambda D} D^{12} dD = \frac{N_0 V}{\lambda^{13}} \int_0^{X_{\max}} e^{-X} X^{12} dX$$

$$= \frac{N_0 V}{\lambda^{13}} \gamma(13, X_{\max}), \quad (16)$$

while the mean total reflectivity factor Z of the volume V is

$$Z = \int_0^{D_{\max}} VN_0 e^{-\lambda D} D^6 dD = \frac{N_0 V}{\lambda^7} \gamma(7, X_{\max}). \quad (17)$$

Thus, the ratio σ_Z/Z is

$$\frac{\sigma_Z}{Z} = \left(\frac{\lambda}{N_0 V} \right)^{\frac{1}{2}} \frac{[\gamma(13, X_{\max})]^{\frac{1}{2}}}{\gamma(7, X_{\max})}. \quad (18)$$

In Fig. 1 we have plotted the ratio σ_Z/Z as a function of R for a volume V of 1 m^3 . For N_0 and the λ - R relation we used the values given by Marshall and Palmer (1948) in Table 1. For a constant rainfall rate R of 10 mm hr^{-1} , we see from Fig. 1 that the measurement of an illuminated volume V of 1 m^3 with a drop camera gives a ratio σ_Z/Z of 40% ; i.e., there is a probability of

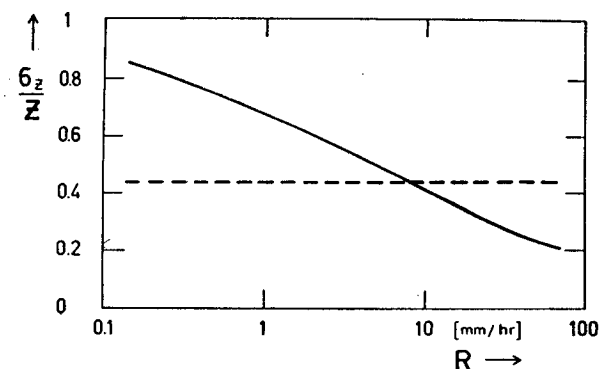


FIG. 1. The ratio σ_Z/Z of the standard deviation σ_Z and the mean reflectivity factor Z as a function of rainfall rate R . The experimental results of Mueller and Sims (1966) are given as a dashed line. The sampling volume V is 1 m^3 .

68% that a measured value of the reflectivity factor deviates less than 40% from the mean Z .

To compare the theoretical results with experiments, we also plotted the σ_Z/Z ratio as measured by Mueller and Sims (1966), i.e.,

$$\frac{\sigma_Z}{Z} = \left(\frac{0.192}{V} \right)^{\frac{1}{2}}, \quad (19)$$

where V is the sampling volume and is measured in

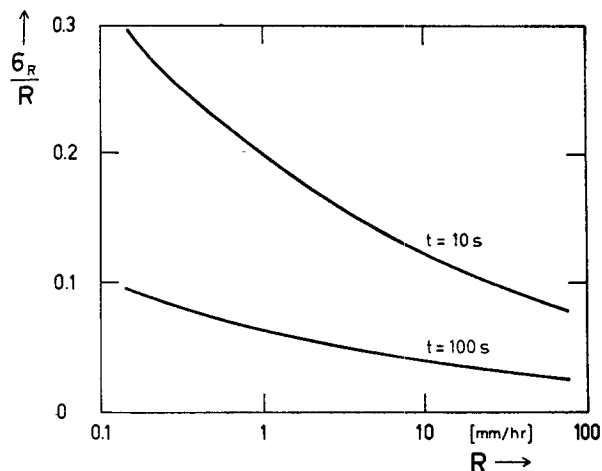


FIG. 2. The ratio σ_R/R of the standard deviation σ_R and the mean rainfall rate R as a function of R . The sampling area is assumed to be $F = 0.01 \text{ m}^2$, the exposure times being 10 sec and 100 sec.

cubic meters. The results of Mueller and Sims are marked with a dashed line in Fig. 1. Their experiments agree well with our theoretical results if we consider that they were not specifically concerned with the dependence of the variance on the rainfall rate R .

In a similar manner to (16) we can also calculate the variance σ_R^2 of the rainfall rate. For filter papers

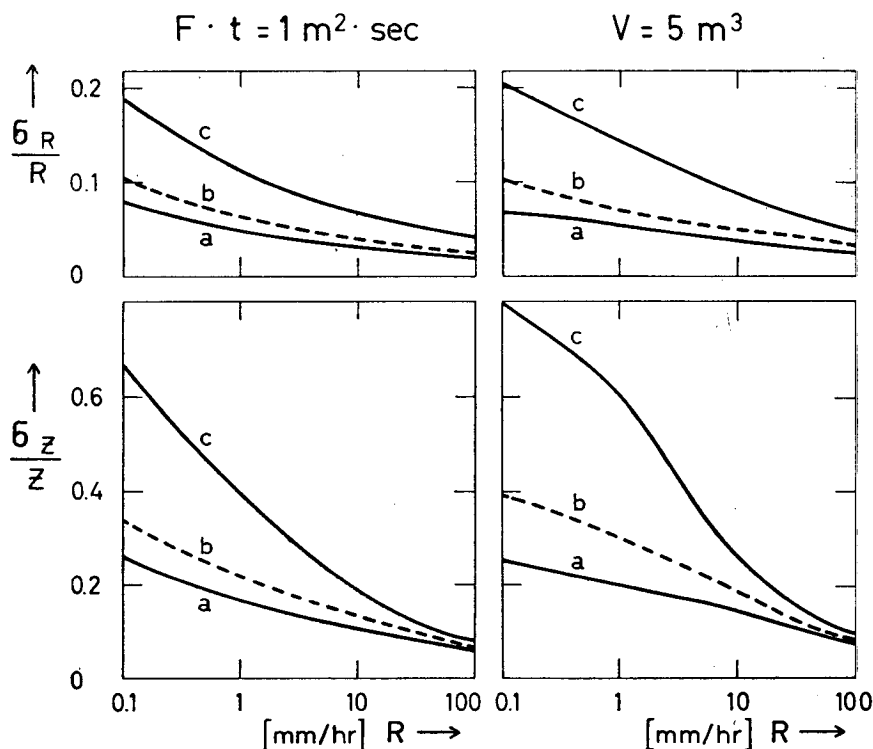


FIG. 3. The ratios σ_R/R and σ_Z/Z as a function of R for the two measurement techniques: filter paper (surface $F \times$ exposing time $t = 1 \text{ m}^2 \text{ sec}$) and drop camera (volume $V = 5 \text{ m}^3$). The indices a, b, c correspond to different types of rainfalls: a, drizzle; b, widespread; c, thunderstorm.

with an area $F=0.01 \text{ m}^2$ during a time $t=100 \text{ sec}$, we get the result plotted in Fig. 2, where again we used the value of N_0 and the λ - R relation given by Marshall and Palmer (1948). For the fall velocity of the raindrops we took the values given by Gunn and Kinzer (1949).

5. Results and conclusions

In Fig. 3 we have plotted the σ_Z/Z and σ_R/R ratios as a function of the rainfall rate R for the two measurement techniques: filter paper (surface $F \times$ exposure time $t=1 \text{ m}^2 \text{ sec}$) and drop camera (volume $V=5 \text{ m}^3$). The ratios have been calculated for three different types of rain given in Table 1. To apply the results plotted in Fig. 3 to different sample sizes, the values have to be multiplied in the case of filter paper (surface F , exposing time t) by

$$\left(\frac{1 \text{ m}^2 \text{ sec}}{Ft}\right)^{\frac{1}{2}} \quad (21)$$

and in the case of the drop camera (illuminated volume V) by

$$\left(\frac{5 \text{ m}^3}{V}\right)^{\frac{1}{2}}. \quad (22)$$

Due to the statistical variation of the drops, we see that a large sample is necessary to get a good estimate of R or Z . For example, in a widespread rain (case b) with a rainfall rate of 1 mm hr^{-1} we need the sample sizes given in Table 2 to find an R or Z value with a probability of 95% which deviates less than 10% from the mean value.

It may be that the requirement for having such a large sample is at least to some extent responsible for the wide variety of Z - R relationships found in the literature.

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TABLE 2. Sample sizes needed to find an R or Z value with a probability of 95% which deviates less than 10% from the mean value, assuming a widespread rain R of 1 mm hr^{-1} .

Measurement technique	R	Z
Ft (filter paper)	$1.5 \text{ m}^2 \text{ sec}$	$19 \text{ m}^2 \text{ sec}$
V (drop camera)	11 m^3	180 m^3

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REFERENCES

- Diem, M., 1966: Rains in the arctic, temperate and tropical zone. Sci. Rept., Meteorologisches Institut, Technische Hochschule, Karlsruhe, Contract DA-91-591-EUC-3634, 93 pp.
- Fujiwara, M., 1965: Raindrop-size distribution from individual storms. *J. Atmos. Sci.*, **22**, 585-591.
- Gunn, R., and G. D. Kinzer, 1949: The terminal velocity of fall for water drops in stagnant air. *J. Meteor.*, **6**, 243-248.
- Hald, A., 1952: *Statistical Theory with Engineering Applications*. New York, Wiley, 717 pp.
- Joss, J., J. C. Thams and A. Waldvogel, 1968: The variation of raindrop size distributions at Locarno. *Proc. Intern. Conf. Cloud Physics*, Toronto, 369-373.
- Lamp, R., 1958: Das Tropfenspektrum in Niederschlägen und die Radar Reflektivität nach fremden und eigenen Messungen. *Beitr. Phys. Atmos.*, **30**, 223-245.
- Marshall, J. S., and W. McK. Palmer, 1948: The distribution of raindrops with size. *J. Meteor.*, **5**, 165-166.
- Mueller, E. A., and A. L. Sims, 1966: The influence of sampling volume on raindrop size spectra. *Proc. Twelfth Conf. Radar Meteor.*, Norman, Okla., Amer. Meteor. Soc., 135-141.
- Sasyo, Y., 1965: On the probabilistic analysis of precipitation particles. *Proc. Intern. Conf. Cloud Physics*, Tokyo, 254-259.
- Stout, G. E., and E. A. Mueller, 1968: Survey of relationships between rainfall rate and radar reflectivity in the measurement of precipitation. *J. Appl. Meteor.*, **7**, 465-474.